

A MULTIVARIATE GENERALIZED BURR-GAMMA DISTRIBUTION

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Summary: A multivariate extension of the Generalized Burr-Gamma family of distributions (Beirlant, de Waal, Teugels, 2000) is defined. This family is applicable in a multivariate analysis setting where extreme values are possible on some or all of the variables and where estimation of extreme tail probabilities is of particular interest. In a sense this model can be used in conjunction with classical extreme value theory (see for instance Tawn, 1988) which concentrates mainly on two-dimensional problems and on the analysis of the tails only.

1. Introduction

During the last few decades several new contributions to bivariate extreme value theory have appeared considering extensions of the univariate threshold methods for extremes to the joint distribution of two variables. Here we can refer to Pickands (1981), Galambos (1987), Resnick (1987), Tawn (1988), Joe (1990) and Joe, Smith and Weissman (1992) among others. The purpose of

this paper is to present a class of distributions that allows the modelling of multivariate data containing extreme values. Extremes do not have to appear on all variables and may not appear on any of the variables. The model is an extension of the univariate generalized Burr-Gamma family of distributions discussed in Beirlant, de Waal and Teugels (2000). In comparison to existing multivariate parametric models, such as the one referred to by Johnson and Kotz (1972) as the multivariate Burr distribution which is defined as a compound Weibull distribution, this model allows for flexible tail modelling comprising all maximal domains of attraction on the margins.

The model will be defined in section 2 and some properties will be considered in section 3. In section 4 we will consider the estimation of the parameters. An illustration on a simulated data set is provided in section 5, while in section 6 we will analyse a real three-dimensional data set. Finally, the model is extended to include covariates in a regression model on the mean.

2. The Multivariate Generalized Burr-Gamma distribution

We will now define the multivariate generalized Burr-Gamma (MGBG) distribution. This is done by imposing that

$$V_{\xi_i} = \frac{1}{\xi_i} \ell n(1 + \xi_i V_i), \quad i = 1, \dots, p \quad (1)$$

are independent $\text{GAM}(k_i, 1)$ random variables for some vector of positive numbers $(k_1, \dots, k_p)$, where

$$V_i = \exp \left\{ \psi(k_i) - \sqrt{\psi'(k_i)} \sum_{(i)}^{-\frac{1}{2}} (\underline{Y} - \underline{\mu}) \right\}, \quad i = 1, \dots, p \quad (2)$$

with

$$\underline{Y} = (Y_1, \dots, Y_p), \quad Y_i = -\ell n X_i, \quad (3)$$

$$\psi(k_i) = \frac{d}{dk_i} \ell n \Gamma(k_i) \quad (\text{the digamma function}),$$

$$\psi'(k_i) = \frac{d}{dk_i} \psi(k_i), \quad (\text{the trigamma function}),$$

$$\underline{\mu} = (\mu_1, \dots, \mu_p)',$$

and $\sum_{(i)}^{-\frac{1}{2}}$ the i -th row of the symmetric square root matrix of Σ^{-1} , where Σ is a symmetric positive definite $p \times p$ matrix.

The joint density function of a MGBG $(\underline{k}, \underline{\mu}, \Sigma, \underline{\xi})$ distributed vector $\underline{X} = (X_1, \dots, X_p)'$ is then given by

$$f(\underline{x}) = \frac{1}{|\Sigma|^{\frac{1}{2}}} \prod_{i=1}^p \frac{\sqrt{\psi'(k_i)}}{\Gamma(k_i) x_i} \exp(-v_{\xi_i}(\underline{x})) v_{\xi_i}(\underline{x})^{k_i-1} \quad (4)$$

$$(1 + \xi_i v_i(\underline{x}))^{-1} v_i(\underline{x}), 1 + \xi_i v_i(\underline{x}) > 0.$$

The parameter space is $\Omega = \{k_i > 0, -\infty < \mu_i < \infty, \Sigma > 0, -\infty < \xi_i < \infty, i = 1, \dots, p\}$. The parameters $\underline{k} = (k_1, \dots, k_p)$ are shape parameters, $\underline{\mu}$ is a vector of location parameters, Σ is a covariance matrix, and $\underline{\xi}$ contains the extreme value indices. Lower case letters y_i, x_i, v_i, v_{ξ_i} denote the values of r.v.'s Y_i, X_i, V_i, V_{ξ_i} . Note that $\underline{V} = (V_1, \dots, V_p)'$ can be written in matrix notation as $\underline{V} = \exp\left\{\psi(\underline{k}) - D_{\psi} \Sigma^{-\frac{1}{2}} (\underline{Y} - \underline{\mu})\right\}$ where $D_{\psi} = \text{diag}\left(\sqrt{\psi'(k_1)}, \sqrt{\psi'(k_2)}, \dots, \sqrt{\psi'(k_p)}\right)$, $\psi(\underline{k}) = (\psi(k_1), \dots, \psi(k_p))'$ and $\exp(\psi(\underline{k})) = (\exp(\psi(k_1)), \exp(\psi(k_2)), \dots, \exp(\psi(k_p)))'$.

3. Special cases

We will now consider some properties and special cases of the MGBG $(\underline{k}, \underline{\mu}, \Sigma, \underline{\xi})$ family of distributions.

(i) Case $\underline{\xi} = \underline{0}$

If $\underline{\xi} = \underline{0}$, the distribution of $\underline{X}$ becomes a multivariate generalized gamma (MGG) with parameters $\underline{k}$, $\underline{\mu}$ and Σ . The distribution of $\underline{Y} = (-\ell n X_1, \dots, -\ell n X_p)'$ will be called a multivariate generalized extreme value (MGEV) distribution, because if $p = 1$, it resembles the well-known type I extreme value $[\text{EV}(\alpha, \beta)]$ distribution with parameters $\alpha = \gamma\beta - \mu$ and $\beta = \frac{\sqrt{6}\sigma}{\pi}$ (see Patil *et al.* (1984), volume 2, p. 54), where γ denotes Euler's constant.

From (2) it follows that

$$\underline{Y} = -\Sigma^{\frac{1}{2}} D_{\psi}^{-1}(\ell n \underline{V} - \psi(\underline{k})) + \underline{\mu}. \quad (5)$$

Furthermore, if $\xi_i = 0$, the variables $V_{\xi_i} = V_i \sim \text{GAM}(k_i, 1)$, $i = 1, \dots, p$, are independent of each other. In this case the marginal distributions are in the maximal domain of attraction of the Gumbel distribution, whence extreme observations are less likely. It follows from (5) that

$$E(\underline{Y}) = \underline{\mu}, \quad \text{Cov}(\underline{Y}, \underline{Y}') = \Sigma. \quad (6)$$

It has been shown in Beirlant *et al.* (2000) that for the univariate case $p = 1$ and $k \rightarrow \infty$, $Y \sim N(\mu, \sigma^2)$. The moment generating function of Y equals

$$\begin{aligned} \phi_Y(s) &= E \exp(sY) \\ &= \exp\left(s\mu + s\sigma\psi(k) / \sqrt{\psi'(k)}\right) \frac{\Gamma(k - s\sigma / \sqrt{\psi'(k)})}{\Gamma(k)}. \end{aligned}$$

Therefore

$$\lim_{k \rightarrow \infty} \phi_Y(s) = \exp\left(s\mu - \frac{1}{2}s^2\sigma^2\right). \quad (7)$$

For the multivariate case, the moment generating function becomes

$$\begin{aligned}
 \phi_Y(\underline{s}) &= E \exp(\underline{s}'Y) \\
 &= \exp \left\{ \underline{s}'\underline{\mu} + \underline{s}'\Sigma^{\frac{1}{2}}\underline{\tau}_k \right\} \prod_{i=1}^p E(V_i^{m_i}) \\
 &= \exp \left\{ \underline{s}'\underline{\mu} + \underline{s}'\Sigma^{\frac{1}{2}}\underline{\tau}_k \right\} \prod_{i=1}^p \frac{\Gamma(k_i + m_i)}{\Gamma(k_i)} \quad (8)
 \end{aligned}$$

$$\text{where } \underline{\tau}_k = \left(\psi(k_i) / \sqrt{\psi'(k_i)}, \dots, \psi(k_p) / \sqrt{\psi'(k_p)} \right)'$$

$$\text{and } m_i = -\underline{s}'\Sigma_{(i)}^{\frac{1}{2}} / \sqrt{\psi'(k_i)}.$$

The moment generating function (8) is defined for $\underline{s}'\Sigma_{(i)}^{\frac{1}{2}} < k_i \sqrt{\psi'(k_i)}$.

It now follows from (7) that

$$\begin{aligned}
 \lim_{\underline{k} \rightarrow \infty} \phi_Y(\underline{s}) &= \exp \left\{ \underline{s}'\underline{\mu} \right\} \prod_{i=1}^p \exp \left\{ -\frac{1}{2} \underline{s}' \Sigma_{(i)}^{\frac{1}{2}} \Sigma_{(i)}^{\frac{1}{2}} \underline{s} \right\} \\
 &= \exp \left\{ \underline{s}'\underline{\mu} - \frac{1}{2} \underline{s}'\Sigma \underline{s} \right\}.
 \end{aligned}$$

This entails that $\lim_{\underline{k} \rightarrow \infty} \text{MGEV}(\underline{k}, \underline{\mu}, \Sigma) = N(\underline{\mu}, \Sigma)$.

(ii) $\underline{\xi} = \underline{0}$ and $\underline{k} = (1, 1, \dots, 1)$

If $\underline{\xi} = \underline{0}$ and $\underline{k}$ is equal to the unitary vector, then the distribution of $\underline{X}$ can be considered as a multivariate Weibull (MWEI) with parameters $\underline{\mu}$ and Σ in the sense that if $p = 1$, $\xi = 0$ and $k = 1$ the MGBG becomes the Weibull with parameters $\alpha = -\gamma\beta - \mu$ and $\beta = \sqrt{6}\sigma/\pi$.

(iii) $\underline{\xi} > \underline{0}$ and $\underline{k} = (1, 1, \dots, 1)$

The case $\xi > 0 (i = 1, \dots, k)$ and $\underline{k} = (1, \dots, 1)$ can be considered as a multivariate Burr case because then the marginals equal univariate Burr distributions. If we condition on the event $V_i > t_i$ for some threshold t_i , $i = 1, \dots, p$, then as $t_i \rightarrow \infty$ the joint survival function of the excesses $Z_i = V_i - t_i (i = 1, \dots, p)$ becomes

$$\begin{aligned} \overline{H}(\underline{z}|\underline{t}) &= P(V_1 > v_1, \dots, V_p > v_p | V_i > t_i, i = 1, \dots, p) \\ &\sim \prod_{i=1}^p \left\{ \frac{\xi_i(z_i + t_i)}{\xi_i t_i} \right\}^{-1/\xi_i} \\ &= \prod_{i=1}^p \left\{ 1 + \frac{\xi_i z_i}{\tilde{\sigma}_i} \right\}^{-1/\xi_i}, \end{aligned} \quad (9)$$

where $\tilde{\sigma}_i = \xi_i t_i$. If $p = 1$, $\overline{H}$ becomes the survival function of the generalized Pareto distribution (GPD). The survival function (9) is a multivariate GPD conditional on $\underline{\mu}$ and Σ known, after transforming $X_1, \dots, X_p$ to $V_1, \dots, V_p$ using (2), and considering the V_i 's exceeding thresholds u_i .

4. Estimating the parameters

In this section we will consider the estimation of the parameters $\underline{k}$, $\underline{\mu}$, Σ and $\underline{\xi}$ from a random sample X_{ij} , $i = 1, \dots, p$; $j = 1, \dots, n$, drawn from a MGBG distribution. Since the extreme value indices ξ_i , $i = 1, \dots, p$, are based on the V_i 's (see (1)), it follows that we can estimate them on the transformed data V_{ij} .

We estimate $\underline{k}$, $\underline{\mu}$ and Σ after trimming the data in order to ignore the extreme values. Let $U = \{x_{ij} : x_{1j} < u_1, x_{2j} < u_2, \dots, x_{pj} < u_p; j = 1, \dots, n\}$ denote the trimmed data set where the u_i denote the trimming limits. Let n_U

denote the sample size of U . The distribution of U can then be approximated by a MGG $(\underline{k}, \underline{\mu}, \Sigma)$. To estimate $\underline{k}$, $\underline{\mu}$ and Σ one can consider the Bayes estimates by simulating the marginal posteriors using the Gibbs sampler. The MDI prior (Zellner (1989)) is easy to derive, namely

$$\pi(\underline{k}, \underline{\mu}, \Sigma) \propto \exp \{I(\underline{k}, \underline{\mu}, \Sigma)\}$$

where

$$I(\underline{k}, \underline{\mu}, \Sigma, \xi) = E \ell n f(\underline{X} | \underline{X} \in U)$$

is the entropy of the distribution, and

$$f(\underline{X} | \underline{X} \in u) = \frac{1}{|\Sigma|^{\frac{1}{2}}} \prod_{i=1}^p \frac{\sqrt{\psi'(k_i)}}{\Gamma(k_i) X_i} \exp(-V_i(\underline{X})) \{V_i(\underline{X})\}^{k_i}.$$

Also, the random variables $V_i(\underline{X}) \approx \text{GAM}(k_i, 1)$, $i = 1, \dots, p$, and these are independent with $E(-\ell n X_i) = \mu_i$.

Therefore

$$I(\underline{k}, \underline{\mu}, \Sigma, \xi) \approx -\frac{1}{2} \ell n |\Sigma| + \sum_{i=1}^p \left\{ \ell n \frac{\sqrt{\psi'(k_i)}}{\Gamma(k_i)} - k_i + \psi(k_i) + \mu_i \right\}. \quad (10)$$

If we take k_i , $i = 1, \dots, p$ as fixed, the MDI prior for $\underline{\mu}$ and Σ becomes

$$\pi(\underline{\mu}, \Sigma) \propto |\Sigma|^{-\frac{1}{2}} \exp \left(\sum_{i=1}^p \mu_i \right). \quad (11)$$

The procedure we adopt here is to estimate $\underline{\mu}$ and Σ by the method of moments, namely

$$\underline{\hat{\mu}} = \underline{\bar{y}} = \sum_{j=1}^{n_U} (y_{1j}, \dots, y_{pj})' / n_U$$

$$\hat{\Sigma} = \left\{ \sum_{j=1}^{n_U} (y_{1j}, \dots, y_{pj})' (y_{1j}, \dots, y_{pj}) - \hat{\underline{\mu}} \hat{\underline{\mu}}' \right\} / n_U, \quad (12)$$

because it provides a quick method and the results do not differ that much from the Bayes estimates. The data can be trimmed visually by looking at a plot of the ordered observations on each variable or by Box plots of the data on each variable. Here we take u_i equal the upper quartile.

Once $\underline{\mu}$ and Σ are estimated, the trimmed data can be transformed to v_{ij} , $i = 1, \dots, p$; $j = 1, \dots, n_U$. The k_i , $i = 1, \dots, p$ can now be obtained by fitting a $\text{GAM}(k_i, 1)$ distribution to each set v_{ij} , $j = 1, \dots, n_U$ using QQ-plots or probability plots.

Estimating ξ_i , $i = 1, \dots, p$

After obtaining $\underline{k}$, we can transform all the data, including the extremes that were deleted to v_{ij} , $i = 1, \dots, p$; $j = 1, \dots, n$, in order to estimate the extreme value indices. Since the independent variables $V_{\xi_i} = \frac{1}{\xi_i} \ell n(1 + \xi_i V_i) \sim \text{GAM}(k_i, 1)$, $i = 1, \dots, p$, the density function of V_i is given by

$$h(v_i) = \frac{1}{\Gamma(k_i)} \exp(-v_{\xi_i}) v_{\xi_i}^{k_i-1} (1 + \xi_i v_i)^{-1}, \quad v_i > 0.$$

Therefore the MDI prior for ξ_i becomes

$$\pi(\xi_i) = \exp\{I(\xi_i)\}$$

where

$$\begin{aligned} I(\xi_i) &= E \ell n h(V_i) \\ &= -\ell n \Gamma(k_i) - k_i + (k_i - 1)\psi(k_i) - \xi_i k_i. \end{aligned}$$

For k_i fixed, the prior becomes

$$\pi(\xi_i) \propto e^{-\xi_i k_i}.$$

If we take $\underline{k} = (1, \dots, 1)$, the Burr case, we can estimate the ξ_i 's using the GPD given in (9) after introducing thresholds t_i , $i = 1, \dots, p$. In general, however, introducing thresholds t_i , $i = 1, \dots, p$, consider

$$G_i = \{v_{ij} : v_{ij} > t_i, j = 1, \dots, n\}, i = 1, \dots, p$$

and let n_{G_i} denote the sample size of G_i . Conditional on G_i for k_i fixed, the likelihood equals

$$\ell(\xi_i | G_i) = \left\{ \frac{1}{\Gamma(k_i)c_i} \right\}^{n_{G_i}} \exp \left\{ - \sum_{j=1}^{n_{G_i}} v_{\xi_i}(j) \right\} \prod_{j=1}^{n_{G_i}} v_{\xi_i}(j)^{k_i-1} \quad (13)$$

where $c_i = \frac{1}{\Gamma(k_i)} \int_{t_i}^{\infty} e^{-u} u^{k_i-1} du$ and $v_{\xi_i}(j) = \frac{1}{\xi_i} \ln(1 + \xi_i v_{ij})$. Therefore the posterior of ξ_i is given by

$$\pi(\xi_i | G_i) \propto \exp(-\xi_i k_i) \ell(\xi_i | G_i). \quad (14)$$

We will illustrate the above estimation procedure in the next section using a simulated data set.

5. A simulated example

To simulate an observation $\underline{x} = (x_1, \dots, x_p)$ from a MGBG($\underline{k}, \underline{\mu}, \Sigma, \underline{\xi}$) distribution, the following steps can be followed:

1. Select $v_{\xi_i} \in \text{GAM}(k_i, 1)$, $i = 1, \dots, p$.
2. Compute $v_i = \frac{1}{\xi_i} \{\exp(\xi_i v_{\xi_i}) - 1\}$, $i = 1, \dots, p$
3. Compute $y_i = -\frac{1}{\sqrt{\psi'(k_i)}} \Sigma_{(i)}^{\frac{1}{2}} \{\ell n \underline{v} - \psi(\underline{k})\} + \mu_i$

$$\begin{aligned} \text{where } \ell n \underline{v} &= (\ell n v_1, \dots, \ell n v_p)' \text{ and} \\ \psi(\underline{k}) &= (\psi(k_1), \dots, \psi(k_p))'. \end{aligned}$$

4. Compute $x_i = \exp(-y_i)$, $i = 1, \dots, p$.

In step 3, $\Sigma_{(i)}^{\frac{1}{2}}$ denotes the i -th row of the symmetric square root matrix of Σ .

Suppose $p = 3$, $\underline{k} = (1, 2, 3)$, $\underline{\mu} = (-3.9, -3.8, -3.6)$

$$\Sigma = \begin{bmatrix} .014 & .004 & .007 \\ .004 & .007 & .004 \\ .007 & .004 & .013 \end{bmatrix} \quad \text{and} \quad \underline{\xi} = \begin{pmatrix} 2.8 \\ 1.4 \\ 0.4 \end{pmatrix}.$$

A sample of $n = 200$ observations was simulated from the MGBG model with the above parameters. The ordered observations on each variable are shown in the graphs of figure 1 with the trimming values $u = (70 \ 65 \ 55)$. These values were chosen as the upper quartiles.

The moment estimates of $\underline{\mu}$ and Σ from the trimmed data are

$$\hat{\underline{\mu}} = \begin{pmatrix} -3.9546 \\ -3.9192 \\ -3.7311 \end{pmatrix} \quad \hat{\Sigma} = \begin{pmatrix} .0225 & .0045 & .0119 \\ .0045 & .0173 & .0041 \\ .0119 & .0041 & .0274 \end{pmatrix}.$$

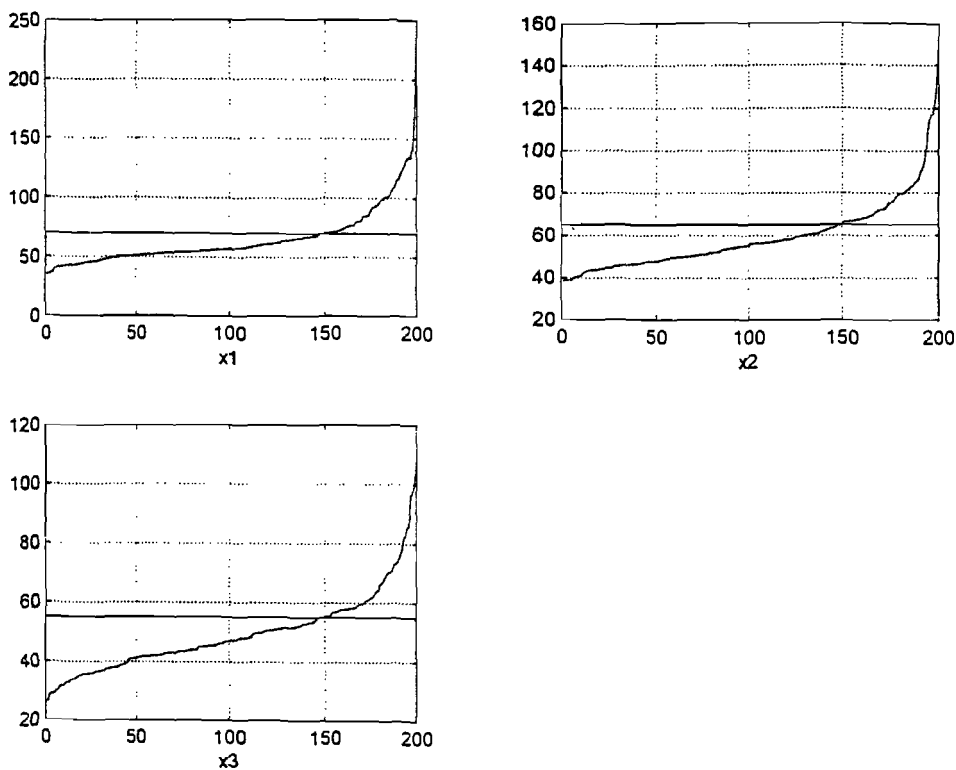


Figure 1: Ordered simulated values on x_1 , x_2 and x_3 with upper quartiles u_1 , u_2 , u_3

The next step is to fit GAM $(k_i, 1)$ distributions, $i = 1, 2, 3$, to the transformed trimmed data v_{ij} . Figure 2 shows the probability plots of the empirical distribution function compared with the GAM $(k_i, 1)$ distribution functions with $k_1 = 2$, $k_2 = 2.5$ and $k_3 = 3$. As a measure of goodness of fit, the divergence measure J (de Waal, 1996) between the empirical distribution function F_n and the fitted GAM $(k_i, 1)$ distribution function F is estimated as $\hat{J}_i = \frac{1}{nU} \sum_{j=1}^{n_U} \ln \frac{F_n(v_{ij})}{F(v_{ij})}$ leading to the values $\hat{J}_1 = 0.0165$, $\hat{J}_2 = 0.0186$, $\hat{J}_3 = 0.0119$. For example, for $k = (1 \ 2 \ 3)$ these measures lead to $\hat{J}_1 = 0.0251$, $\hat{J}_2 = 0.0210$ and $\hat{J}_3 = 0.0119$. The closer they get to zero, the better the fit.

Judging these fits to be reasonable we proceeded to estimate the extreme value indices ξ_1 , ξ_2 and ξ_3 . Taking thresholds $\underline{t} = (12 \ 13 \ 14)$ on the v_{ij} 's ($i = 1, 2, 3$), respectively after transforming all the data to v 's, we simulated values ξ_i using (14). The marginal posteriors function values at the simulated values of ξ_i are shown in figure 3. Taking the posterior means as the estimates we obtain

$$\hat{\xi}_1 = 1.0808, \quad \hat{\xi}_2 = 0.7420, \quad \hat{\xi}_3 = 0.3161.$$

The 90% hpd regions on these parameters are $(0.8631, 1.3910)$, $(0.5704, 0.9507)$ and $(0.1796, 0.5125)$ respectively. The ξ values are strongly underestimated, but when drawing parametric bootstrap samples using these estimates and comparing the bootstrap data with the original data through QQ-plots (see figure 4) fairly close relationships are observed.

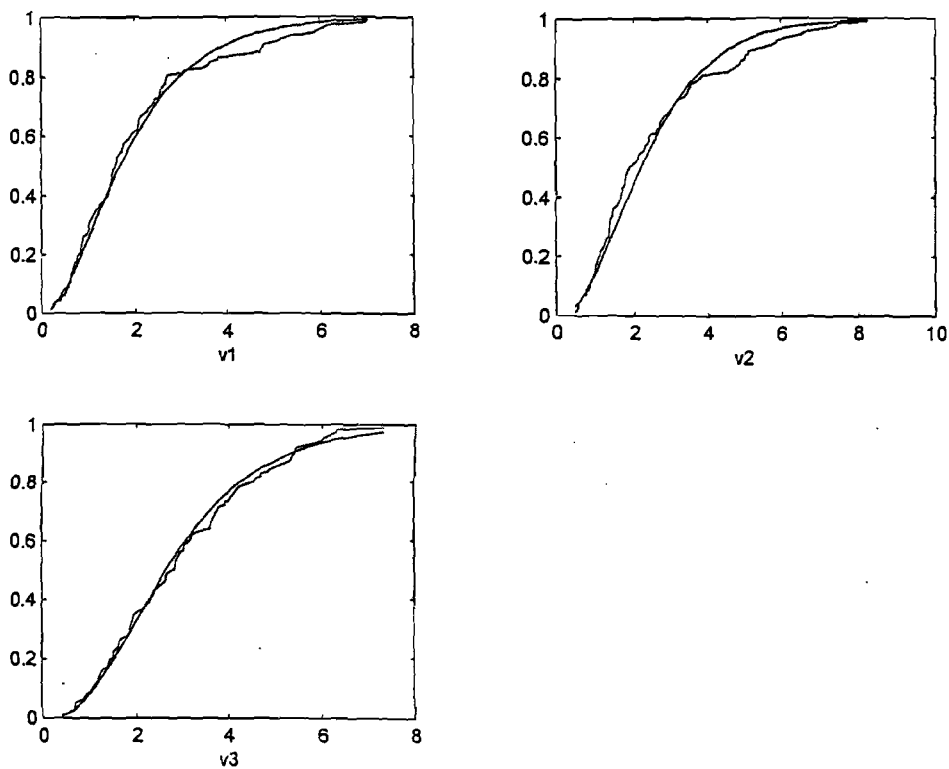


Figure 2. Probability plots of the empirical cdf compared with cdf of GAM $(k_i, 1)$ distributions for $k_1 = 2$, $k_2 = 2.5$ and $k_3 = 3$

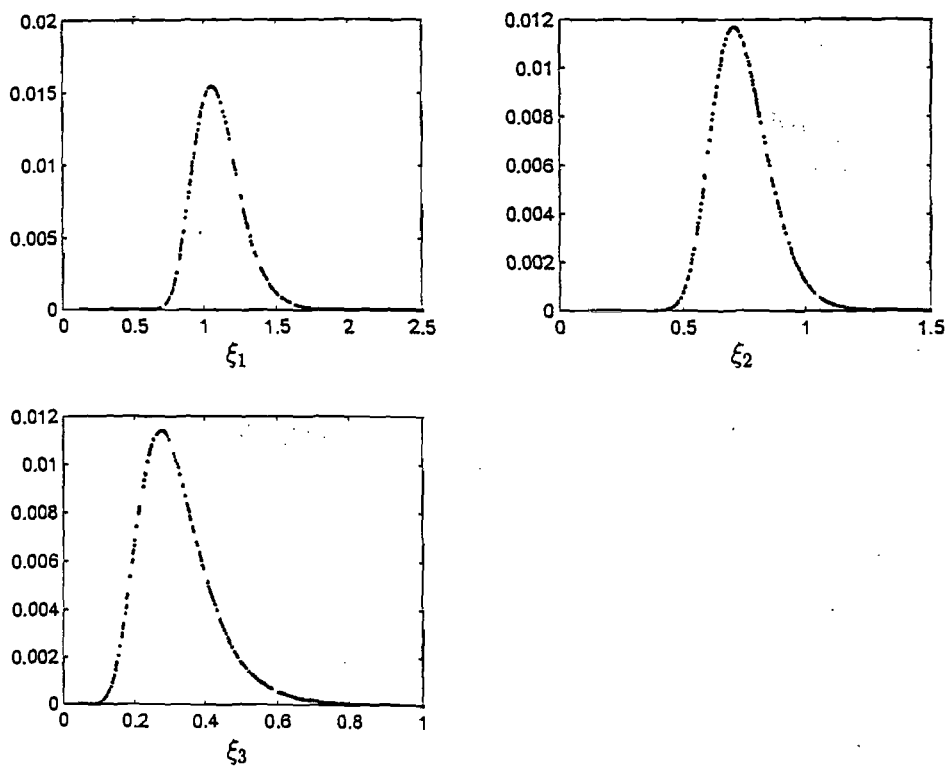


Figure 3. Posterior distributions of ξ_i , $i = 1, 2, 3$

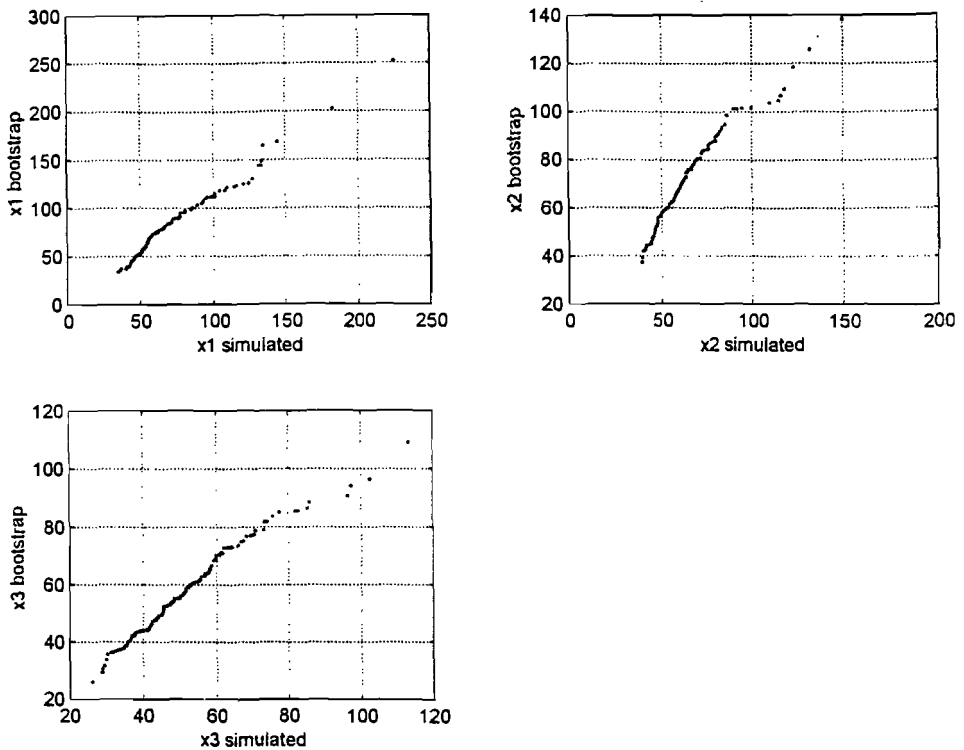


Figure 4. QQ-plots on each variable comparing the original simulated data with the bootstrap data

6. An application to wind speed modelling with regression

The maximum monthly wind speed together with the wind directions are recorded at three stations in the Cape Town area, namely Table Bay (TB), Airport (AP) and Robben Island (RI). In the appendix the data for the period from March 1993 to December 1998 are given. Let $\underline{X} = (X_{TB}, X_{AP}, X_{RI})$ denote the wind speed in knots and $\underline{\theta} = (\theta_{TB}, \theta_{AP}, \theta_{RI})$ the corresponding wind direction in radials. The data are graphed in figure 5.

Assuming the MGBG model, the estimates of $\underline{\mu}$ and $\underline{\Sigma}$ were calculated as

$$\hat{\underline{\mu}} = \begin{pmatrix} -3.8840 \\ -3.8388 \\ -3.5979 \end{pmatrix} \quad \hat{\underline{\Sigma}} = \begin{pmatrix} .0140 & .0038 & .0072 \\ .0038 & .0070 & .0041 \\ .0072 & .0041 & .0133 \end{pmatrix}.$$

after trimming the data at $\underline{u} = (60, 55, 45)$. The values for $\underline{k} = (0.8, 1.0, 1.5)$ were chosen. The thresholds $\underline{t} = (0.8, 1.2, 1.3)$ were used on the v 's. The estimates for $\underline{\xi}$ are given by $\underline{\xi} = (2.8867, 1.3645, 0.3997)$.

Introducing the angle of the wind gusts as covariates, we introduce the model

$$\underline{\mu} = \underline{\beta}_0 + \underline{\beta}_1 \theta_{TB} + \underline{\beta}_2 \theta_{AP} + \underline{\beta}_3 \theta_{RI} \quad (15)$$

where $\theta = \begin{cases} 0 & \text{if the gust is from the North} \\ 1 & \text{if the gust is from the South.} \end{cases}$

Ordinary least squares estimates of $\underline{\beta} = (\underline{\beta}_0 \ \underline{\beta}_1 \ \underline{\beta}_2 \ \underline{\beta}_3)$, from the regression of $\underline{y} = -\ln \underline{x}$ on θ_{TB} , θ_{AP} , θ_{RI} on the trimmed data at $\underline{u}$ as given above, are

$$\hat{\underline{\beta}} = \begin{pmatrix} -3.7798 & 0.0082 & -0.0779 & -0.0751 \\ -3.8826 & 0.0029 & 0.0309 & -0.0429 \\ -3.4770 & 0.0256 & -0.0330 & -0.1412 \end{pmatrix}.$$

the corresponding wind direction in radials. The data are graphed in Figure 5.

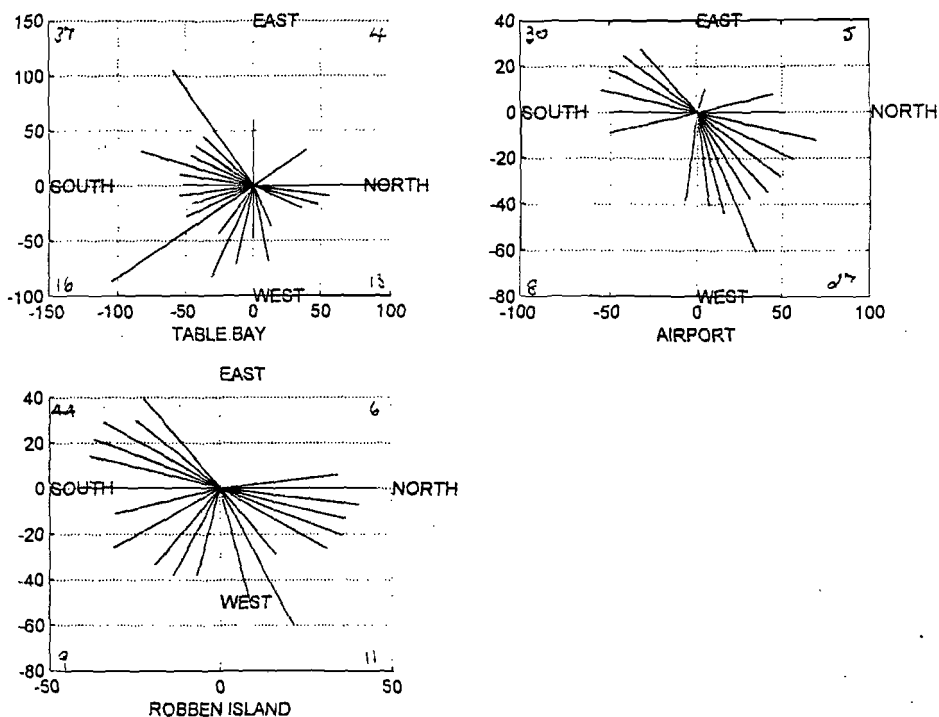


Figure 5. Maximum monthly wind speeds and directions measured at Table Bay, Airport and Robben Island during March 1993–December 1998

The Bayes estimates of the extreme value indices using the thresholds t as given above, now become $\hat{\xi}_1 = 3.371$, $\hat{\xi}_2 = 1.2149$, $\hat{\xi}_3 = 0.5174$. From a parametric bootstrap sample of size 5000 using these estimates, 0.14% of the generated values were found within the region $\{X_{TB} > 100, X_{AP} > 70, X_{RI} > 50\}$ given the wind is from the north at all three stations. This probability becomes three times larger if the wind direction is from the south. Under the reduced model, ignoring the wind direction, i.e. setting $\underline{\beta}_1 = \underline{0}$, $\underline{\beta}_2 = \underline{0}$, $\underline{\beta}_3 = \underline{0}$, this proportion was found to be 0.14%.

7. Conclusion

We defined a multivariate model that covers a family of distributions which can contain extremes. It is argued that, next to central parts of a multivariate data set, also multivariate extremes can be modelled by the given parametric MGBG model, in a way which is analogous to classical peaks over threshold methods. A fully Bayesian approach to the estimation of *all* parameters will be the subject of a future communication. Such a procedure would have the advantage of finding hpd regions for these parameters and tail probabilities.

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APPENDIX

Monthly maximum wind gusts (Directions and Strengths) measured (Radials and Knots) at Cape Town Harbour, Cape Town Airport and Robben Island during March 1993 to December 1998.

HARBOUR		AIRPORT		ROBBEN ISLAND	
Angle	Strength	Angle	Strength	Angle	Strength
2.0944	121.0000	2.9671	45.0000	2.4435	35.0000
3.4907	49.0000	2.9671	56.0000	2.6180	37.0000
5.0615	39.0000	4.8869	41.0000	5.2360	27.0000
6.1087	41.0000	5.2360	45.0000	0.1745	35.0000
6.2832	56.0000	5.9341	60.0000	6.2832	39.0000
6.1087	56.0000	5.9341	47.0000	5.7596	37.0000
2.6180	39.0000	5.7596	49.0000	2.7925	41.0000
2.7925	49.0000	2.7925	43.0000	2.2689	37.0000
2.4435	53.0000	2.9671	47.0000	2.6180	39.0000
2.4435	53.0000	3.1416	49.0000	2.6180	43.0000
2.6180	51.0000	2.9671	47.0000	2.4435	39.0000
2.2689	51.0000	2.9671	47.0000	2.6180	39.0000
3.1416	47.0000	2.9671	43.0000	2.4435	39.0000
1.5708	60.0000	5.7596	45.0000	2.2689	37.0000
4.3633	45.0000	6.1087	45.0000	4.5379	37.0000
4.3633	89.0000	5.2360	70.0000	4.8869	53.0000
6.1087	49.0000	6.1087	54.0000	5.9341	39.0000
2.7925	43.0000	2.6180	45.0000	2.6180	43.0000
5.7596	41.0000	5.4105	45.0000	2.6180	33.0000
3.8397	43.0000	2.9671	39.0000	2.4435	31.0000

HARBOUR		AIRPORT		ROBBEN ISLAND	
Angle	Strength	Angle	Strength	Angle	Strength
2.6180	43.0000	2.9671	45.0000	2.6180	33.0000
2.6180	47.0000	2.6180	47.0000	2.6180	37.0000
2.4435	56.0000	2.9671	51.0000	2.7925	37.0000
2.6180	49.0000	2.9671	43.0000	2.6180	35.0000
3.8397	62.0000	2.7925	41.0000	2.7925	35.0000
0.6981	51.0000	2.6180	47.0000	5.0615	64.0000
3.8397	136.0000	5.5851	51.0000	3.4907	33.0000
4.8869	70.0000	5.7596	56.0000	5.0615	45.0000
2.7925	89.0000	6.2832	51.0000	6.2832	33.0000
4.8869	58.0000	2.7925	45.0000	3.8397	41.0000
5.7596	41.0000	5.9341	45.0000	0.1745	31.0000
2.7925	54.0000	5.5851	54.0000	2.4435	45.0000
2.6180	45.0000	2.7925	47.0000	2.0944	37.0000
2.7925	54.0000	2.7925	53.0000	2.4435	41.0000
2.6180	54.0000	2.6180	49.0000	2.4435	39.0000
2.4435	49.0000	3.3161	51.0000	2.4435	31.0000
2.7925	53.0000	2.6180	47.0000	2.6180	35.0000
2.4435	49.0000	2.7925	47.0000	2.0944	31.0000
2.4435	37.0000	0.1745	45.0000	6.2832	29.0000
5.9341	51.0000	6.1087	64.0000	5.7596	41.0000
4.7124	47.0000	6.1087	49.0000	5.2360	33.0000
6.2832	97.0000	6.2832	49.0000	6.1087	41.0000
4.7124	47.0000	6.1087	53.0000	4.1888	37.0000
3.3161	56.0000	2.9671	47.0000	2.6180	41.0000
2.6180	53.0000	2.7925	51.0000	3.1416	41.0000

HARBOUR		AIRPORT		ROBBEN ISLAND	
Angle	Strength	Angle	Strength	Angle	Strength
2.6180	51.0000	5.9341	53.0000	2.7925	37.0000
2.2689	53.0000	2.9671	49.0000	2.7925	37.0000
2.2689	58.0000	3.1416	51.0000	2.4435	39.0000
2.6180	54.0000	5.5851	45.0000	2.7925	37.0000
3.8397	47.0000	4.5379	39.0000	4.3633	41.0000
4.1888	49.0000	6.2832	47.0000	2.4435	31.0000
3.6652	58.0000	6.1087	70.0000	6.2832	45.0000
2.9671	37.0000	6.1087	43.0000	2.4435	33.0000
4.5379	72.0000	6.2832	51.0000	5.5851	41.0000
2.4435	53.0000	3.1416	45.0000	2.0944	45.0000
2.7925	45.0000	2.7925	45.0000	2.4435	37.0000
2.4435	43.0000	5.4105	49.0000	2.2689	37.0000
2.9671	56.0000	2.7925	54.0000	2.6180	41.0000
2.9671	54.0000	2.7925	51.0000	2.2689	39.0000
2.9671	54.0000	3.1416	49.0000	2.2689	39.0000
2.9671	49.0000	2.7925	45.0000	2.7925	33.0000
3.1416	47.0000	3.3161	37.0000	2.2689	29.0000
4.1888	51.0000	5.5851	47.0000	4.5379	39.0000
6.1087	43.0000	6.1087	41.0000	5.5851	29.0000
6.2832	43.0000	6.1087	47.0000	4.1888	39.0000
4.7124	45.0000	5.0615	47.0000	4.1888	35.0000
3.1416	51.0000	6.1087	47.0000	2.0944	33.0000
2.6180	49.0000	2.7925	41.0000	2.0944	37.0000
3.3161	53.0000	2.4435	43.0000	2.4435	39.0000
2.7925	54.0000	3.1416	43.0000	2.0944	39.0000